

Time-Resolved Optically Detected Magnetic Resonance of Nitrogen-Vacancy Centers in Diamond

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This report details the implementation and results of Time-Resolved Optically Detected Magnetic Resonance (TR-ODMR) measurements on Nitrogen-Vacancy (NV) centers in a bulk diamond sample. NV centers have a long coherence time, and can be initialized and measured optically. We performed Continuous Wave (CW) ODMR to characterize zero-field splitting and magnetic field response. Time-resolved measurements were conducted to observe Rabi oscillations and determine the longitudinal relaxation time T_1 .

I. INTRODUCTION

The negatively charged nitrogen-vacancy center (NV^-) in diamond is a point defect formed by a substitutional nitrogen atom next to a vacancy. Its electronic ground state is a spin triplet ($S = 1$) with a zero-field splitting between the $m_s = 0$ and $m_s = \pm 1$ sublevels of approximately $D \approx 2.87$ GHz at room temperature. In the effective spin-1 description used in this work, the ground-state Hamiltonian can be written as

$$\frac{H}{h} = DS_z^2 + E(S_x^2 - S_y^2) + \gamma_e \mathbf{B} \cdot \mathbf{S}, \quad (1)$$

[1] where E describes strain- and electric-field-induced splitting, γ_e is the electron gyromagnetic ratio, and $\mathbf{B}$ is the applied magnetic field.

ODMR combines optical spin polarization/readout with microwave-driven magnetic resonance. Under green excitation, spin-selective intersystem crossing pumps the NV ensemble preferentially into $m_s = 0$, and the fluorescence is higher for $m_s = 0$ than for $m_s = \pm 1$. When the microwave drive is resonant with a spin transition, population is transferred out of $m_s = 0$ and a dip appears in fluorescence. In zero magnetic field this produces the characteristic two-peak structure around D due to nonzero E .

After CW frequency calibration, we used time-resolved pulse sequences to probe longitudinal relaxation and coherent control. The T_1 sequence measures recovery toward thermal equilibrium as a function of delay τ , while the Rabi represent coherent spin rotations driven by resonant microwave pulses of variable duration. The measured signals were fitted with the models

$$V_{T_1}(\tau) = A \exp(-\tau/T_1) + C, \quad (2)$$

and

$$V_R(\tau) = A \exp \left[- \left(\tau/T_2^{\text{eff}} \right)^\beta \right] \sin(\omega_{\text{Rabi}}\tau + \phi) + B\tau + C. \quad (3)$$

II. RESULTS AND DISCUSSION

A. Photoluminescence and Lock-in Calibration

We carried out our measurements in workstation 1 using the D9 sample. During setup alignment, the detector background level before laser excitation was approximately 0.22 V, while

the best photoluminescence operating point reached about 1.5 V. The sample position was adjusted to find the region of highest density of nitrogen vacancy sites by maximizing photoluminescence signal at a voltmeter.

For the ODMR measurements we slightly re-calibrated the sample position to optimize for the lock-in signal, which measures the difference in photoluminescence between the microwave excitation time and the non-microwave excitation time. The sample's concentration of nitrogen vacancies and the microwave's intensity are inhomogeneous in space. We find the position that with both the highest concentration of nitrogen vacancies and highest intensity of the microwave field. We could reach a maximum signal level of 18.5 mV, with a noise floor of 0.1 mV. We used a lock-in time constant of 10 ms.

B. CW ODMR and Magnetic Field Measurements

Initially, a frequency sweep was conducted in zero magnetic field to characterize the ground-state splitting of the NV centers (Fig. 1). The resonance exhibits a double-peak structure due to the strain-induced crystal field parameter E .

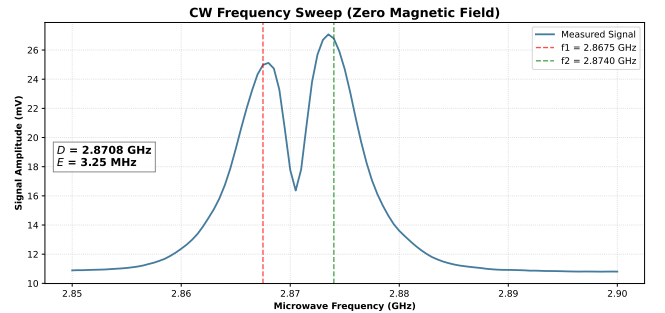


FIG. 1: CW ODMR sweep at zero magnetic field from 2.3 to 3.4 GHz range with extracted D and E parameters indicated by the fit.

By analyzing this splitting, we extracted $D = 2.87$ GHz and the splitting parameter $E = 3.25$ GHz. The obtained D value is consistent with the standard literature value for NV^- centers at room temperature.

In the zero-field scan, two resonance peaks are observed, corresponding to the $m_s = 0 \rightarrow m_s = +1$ and $m_s = 0 \rightarrow m_s = -1$

–1 transitions split by E . D is the average of the two resonance frequencies and E is half of their separation.

After enabling microwave modulation and lock-in detection, the optimization target became fluorescence contrast between microwave-on and microwave-off conditions. The best lock-in amplitude was approximately $R \approx 18.5$ mV with a noise floor of about 0.1 mV. This behavior is consistent with spatial inhomogeneity in both NV density and microwave field strength across the sample and coplanar waveguide.

We apply a arbitrary direction magnetic field and re-do the sweep as can be seen in Fig. 2(a). With an external magnetic field, the spectrum is expected to show up to eight resonances: four crystallographic NV orientations, each with two allowed transitions. In our data, the observed spectrum is consistent with this orientation-split ODMR picture, although 6 peaks only can be clearly seen. A focused sweep of one peak (Fig. 2(b)) was then used to identify the working resonance frequency at 3.0912 GHz, which was used for the time-resolved measurements. We used a lock-in time constant of 30 ms to improve stability.

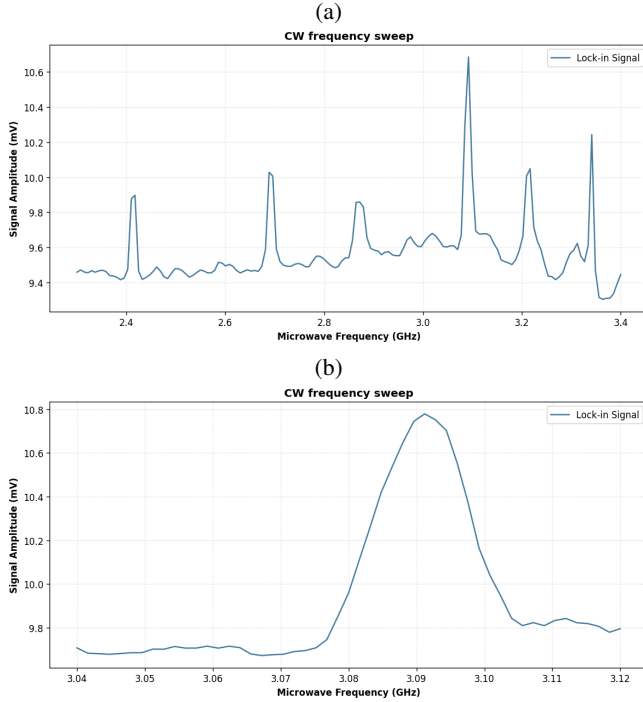


FIG. 2: CW ODMR spectra under magnetic field: (a) full sweep from 2.3 to 3.4 GHz; (b) high-resolution sweep from 3.04 to 3.12 GHz focused on the selected resonance at 3.0912 GHz.

C. Rabi Oscillations

Coherent control of the NV spin was investigated using the Rabi oscillation pulse sequence. Our group's experimental results are shown in Fig. 3. While clear sinusoidal oscillations were limited in our physical setup, the transition to saturation and the characteristic damping times were successfully cap-

tured by the fitting model. The fitting was done according to the formula:

$$V_R(\tau) = Ae^{-(\tau/T_2^{\text{eff}})^\beta} \sin(\omega_{\text{Rabi}}\tau + \phi) + B\tau + C$$

T_2^{eff} : The effective relaxation time characterizing the dephasing timescale of the NV center ensemble.

β : A dimensionless empirical parameter that accounts for the inhomogeneous sample and MW field. [2]

ω_{Rabi} : The Rabi frequency, the angular rate at which the spin state is driven between the $|0\rangle$ and $|1\rangle$ states. It is directly proportional to the amplitude of the applied microwave magnetic field.

B : A coefficient introduced to model temperature-dependent baseline drift during the measurement due to the MW power.

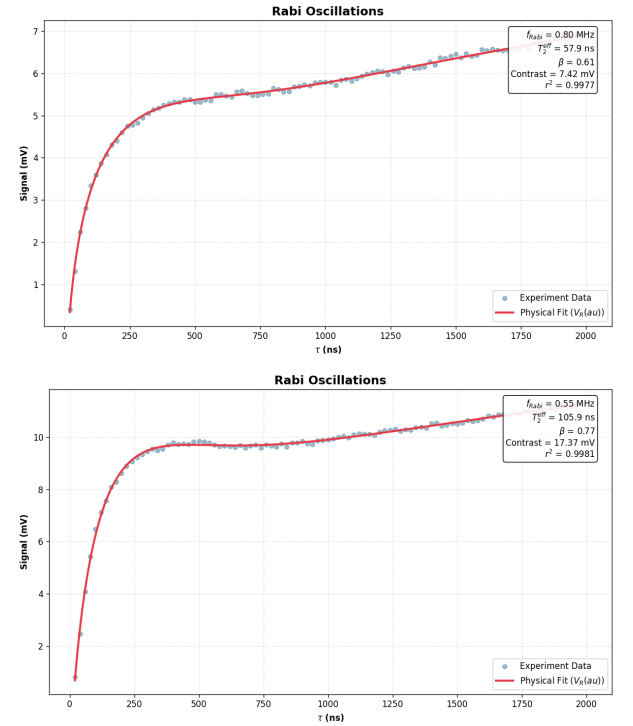
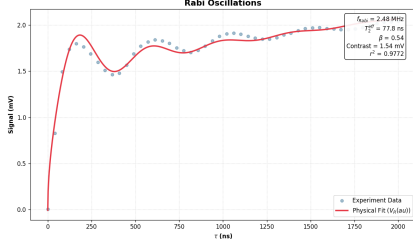


FIG. 3: Rabi oscillation measurements performed by our group: (a) worse calibrated setup, (b) better calibrated setup.

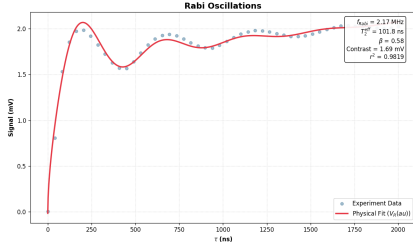
In Fig. 3 (a) we could not get a large enough signal amplitude. Then after re-calibrating the setup by adjusting the sample and magnet positions, we obtained a better measurement Fig. 3 (b). While we still couldn't properly measure ω_{Rabi} , we could accurately measure T_2^{eff} to be around 100 ns, which is close to the value measured by our colleagues and shown in Fig. 4. Unfortunately we were not able to reliably measure the Rabi oscillations. We believed that the main issue was the low signal, but even after optimizing for a high signal, we still could not get a good reading of the ω_{Rabi} . This led us to believe that the MW excitation was not sufficient to observe the Rabi oscillations, and unfortunately we could not

move the MW antenna or calibrate it to achieve the desired phenomena. [3]

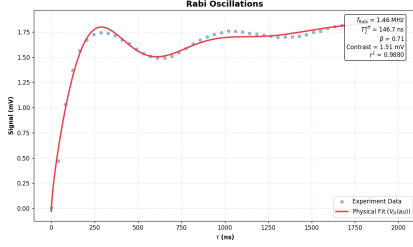
We then analyzed four reference measurements kindly provided by colleagues Volk János and Eisenberger Lea (Fig. 4) to compare with our fitting model.



(a) No attenuation



(b) 6 dB attenuation



(c) 10 dB attenuation

FIG. 4: Colleague-provided Rabi oscillation measurements showing coherent nutations at various power levels.

The ω_{Rabi} measurement with different MW attenuations allows us to look at the power dependence of the frequency. We expect $\omega_{\text{Rabi}} \propto \sqrt{P}$:

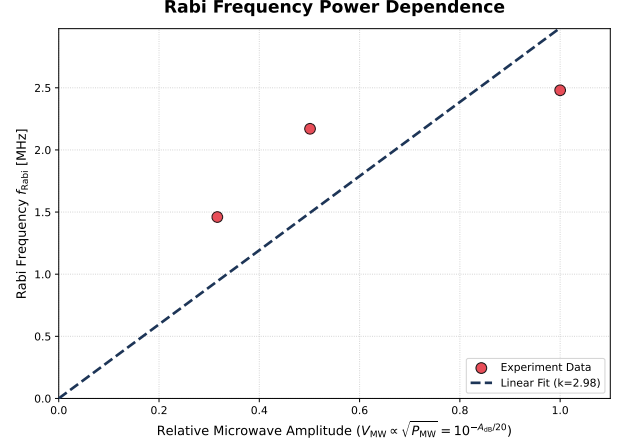


FIG. 5: Power dependence of the Rabi frequency.

As expected, ω_{Rabi} increases with the power. The few data points and the non ideality of the MW and amplifier explain the deviation from the linear fit. On the other hand T_2^{eff} decreases with an increasing power. This is mainly due to the effect of the inhomogeneous MW field, which creates a spread of ω_{Rabi} , which macroscopically destructively interfere making the dephasing quicker. This is in accordance with the measurements.

D. T_1 Relaxation Time

The longitudinal relaxation time T_1 was measured using a variable-delay sequence. Figure 6 compares our group's cleanest experimental dataset with the reference sweep provided by the laboratory instructor. The signal follows an exponential decay, $V(\tau) = A \exp(-\tau/T_1) + C$.

The sequence design followed the standard timing constraints for this setup, with long initialization and short read-out pulses (nominally on the order of $t_{\text{init}} \sim 50 \mu\text{s}$ and $t_{\text{read}} \sim 10 \mu\text{s}$). The polarized state yields a higher fluorescence level than the relaxed state, giving the positive contrast required for T_1 fitting.

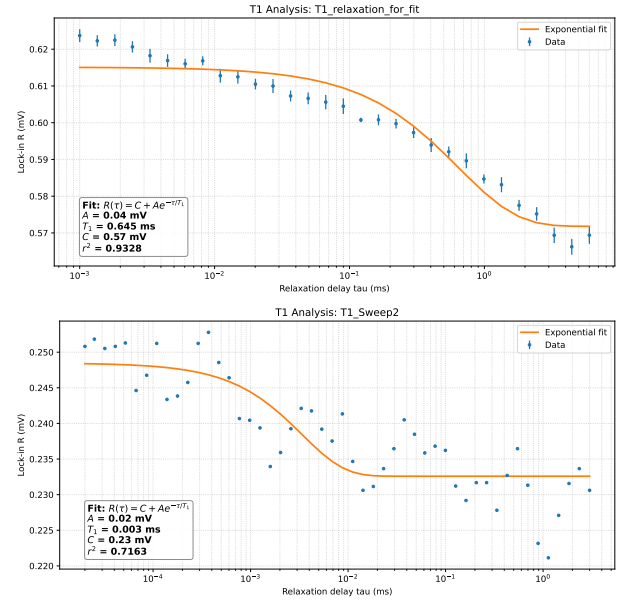


FIG. 6: T1 relaxation measurements. (Top) Instructor reference data; (Bottom) Our group data.

Our measured data was very noisy, and a more careful measurement setup is needed for a more precise analysis. The relaxation time obtained by fitting Fig. 6 (Top) is 645 ns, orders of magnitude larger than T_2^{eff} measured in the Rabi oscillation measurement. This is as expected because T1 involves longitudinal energy relaxation, while T_2^{eff} involves transverse decoherence.

III. CONCLUSION

We successfully implemented ODMR characterization of NV centers in diamond. We extracted the zero-field splitting D and E , measuring the magnetic field response, and identifying the optimal modulation frequency at 3.0912 GHz. We analyzed the relaxation time of the laser excitation, and MW driven Rabi oscillations. Our results, validated against colleague and instructor data, provide a clear quantification of the coherent nutation and relaxation regimes of the system.

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- [1] D. Lönard, I. C. Barbosa, S. Johansson, J. Gutsche, and A. Widera, Limits of absolute vector magnetometry with nv centers in diamond (2025), arXiv:2504.20750 [quant-ph].
 - [2] There is an ensemble of NV centers, and each has a slightly different environment and Rabi frequency because of the inhomogeneity. This parameter accounts for the spread of the distribution of frequencies, and is less than 1. In an ideal case it would

be 1.

- [3] During the measurement exercise, we thought there was no point in doing a power-dependence analysis using an attenuator since we couldn't even measure the Rabi oscillations, but we later realized we could've done a power-dependence study of the T2 using our measurement setup, and that can give us information about the dependence of the rabi frequency as well.